Two-stage Gauss–Legendre Method

Exact one-step Jacobian, symplecticity, and symmetry

Fourth-order collocation method for canonical Hamiltonian systems

1 Hamiltonian convention

Let $z \in \mathbb{R}^{2d}$ and consider the possibly time-dependent canonical Hamiltonian system

$$\dot{z} = f(t,z) = \Omega^{-1}\nabla_z H(t,z), \qquad \Omega = \begin{pmatrix} 0 & I_d \\ -I_d & 0 \end{pmatrix}. \tag{1}$$

For the component-major guiding-centre state

$$z = (x_1,\dots,x_d,y_1,\dots,y_d)^T,$$

this convention gives

$$f(t,z) = \begin{pmatrix} -\nabla_y H(t,z) \\ \nabla_x H(t,z) \end{pmatrix}.$$

At any fixed time and state, let $F = D_z f(t,z)$. Since $\nabla_z^2 H$ is symmetric,

$$F^T\Omega + \Omega F = 0. \tag{2}$$

This identity is the infinitesimal Hamiltonian condition used below. The proof remains valid for a time-dependent Hamiltonian because the stage times do not depend on the initial state z_n.

2 Gauss–Legendre stages

Set

$$r = \frac{\sqrt{3}}{6}, \qquad c_1 = \frac{1}{2} - r, \qquad c_2 = \frac{1}{2} + r.$$

The Butcher coefficients of the two-stage Gauss–Legendre method are

$$A = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} - r \\ \frac{1}{4} + r & \frac{1}{4} \end{pmatrix}, \qquad b = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}. \tag{3}$$

For one step of length h, the stage states satisfy

$$Z_i = z_n + h\sum_{j=1}^{2} a_{ij} f(t_n + c_j h, Z_j), \qquad i = 1,2, \tag{4}$$

and the accepted state is

$$z_{n+1} = \Phi_{h,t_n}(z_n) = z_n + h\sum_{i=1}^{2} b_i f(t_n + c_i h, Z_i). \tag{5}$$

The method is implicit: equations (4) determine (Z_1, Z_2) simultaneously.

3 Direct proof of fourth-order accuracy

We now prove directly why the Runge–Kutta order conditions through degree four give a fourth-order method. For clarity, first consider the autonomous problem

$$\dot{y} = f(y), \qquad y(t_n) = y. \tag{6}$$

The non-autonomous case follows by augmenting the state with time, $\widehat{y} = (t, y)$ and $\dot{\widehat{y}} = (1, f(t, y))$. Because $c = A\mathbf{1}$, applying the Runge–Kutta method to this augmented autonomous system produces exactly the stage times used in (4).

Set

$$F = f(y), \qquad F' = Df(y), \qquad F'' = D^2 f(y), \qquad F''' = D^3 f(y), \tag{7}$$

where the derivatives are multilinear maps. Repeated application of the chain rule gives

$$\begin{aligned} y' &= F, \\ y'' &= F'F, \\ y''' &= F''(F, F) + F'^2 F, \\ y^{(4)} &= F'''(F, F, F) + 3F''(F, F'F) + F'F''(F, F) + F'^3 F. \end{aligned} \tag{8}$$

Consequently, the exact flow has the expansion

$$\begin{aligned} y(t_n + h) = y + hF &+ \frac{h^2}{2} F'F \\ &+ \frac{h^3}{6} \left[F''(F, F) + F'^2 F \right] \\ &+ \frac{h^4}{24} \left[F'''(F, F, F) + 3F''(F, F'F) + F'F''(F, F) + F'^3 F \right] + O(h^5). \end{aligned} \tag{9}$$

For a generic Runge–Kutta step, write

$$K_i = f(Y_i), \qquad Y_i = y + h \sum_j a_{ij} K_j, \qquad y_{n+1} = y + h \sum_i b_i K_i, \tag{10}$$

and define $C = \operatorname{diag}(c)$. Powers such as c^2 and c^3 are understood componentwise. If $\Delta_i = h \sum_j a_{ij} K_j$, Taylor expansion about y gives

$$K_i = F + F'\Delta_i + \frac{1}{2} F''(\Delta_i, \Delta_i) + \frac{1}{6} F'''(\Delta_i, \Delta_i, \Delta_i) + O(h^4). \tag{11}$$

Substituting the stage expansions recursively into (11) and collecting equal powers of h yields

$$\begin{aligned} K_i = F &+ hc_i F'F \\ &+ h^2 \left[(Ac)_i F'^2 F + \frac{1}{2} c_i^2 F''(F, F) \right] \\ &+ h^3 \Bigg[(A^2 c)_i F'^3 F + \frac{1}{2} (Ac^2)_i F'F''(F, F) \\ &\qquad + c_i (Ac)_i F''(F, F'F) + \frac{1}{6} c_i^3 F'''(F, F, F) \Bigg] + O(h^4). \end{aligned} \tag{12}$$

Inserting (12) into the accepted update in (10) gives

$$\begin{aligned} y_{n+1} = y &+ h(b^T \mathbf{1})F + h^2 (b^T c) F'F \\ &+ h^3 \left[(b^T Ac) F'^2 F + \frac{1}{2} (b^T c^2) F''(F, F) \right] \end{aligned}$$

$$+ h^4 \Big[(b^T A^2 c) F'^3 F + \frac{1}{2}(b^T A c^2) F' F''(F,F) \\ + (b^T C A c) F''(F, F'F) + \frac{1}{6}(b^T c^3) F'''(F,F,F) \Big] + O(h^5). \tag{13}$$

The elementary differentials in (9) and (13) are independent for a general smooth vector field. Their coefficients agree through h^4 precisely when

$$\begin{aligned} b^T \mathbf{1} &= 1, & b^T c &= \frac{1}{2}, \\ b^T c^2 &= \frac{1}{3}, & b^T A c &= \frac{1}{6}, \\ b^T c^3 &= \frac{1}{4}, & b^T C A c &= \frac{1}{8}, \\ b^T A c^2 &= \frac{1}{12}, & b^T A^2 c &= \frac{1}{24}. \end{aligned} \tag{14}$$

For example, the coefficient of $F''(F,F)$ in the numerical step is $\frac{1}{2} b^T c^2$ and its exact coefficient is $\frac{1}{6}$, which gives $b^T c^2 = \frac{1}{3}$. The other seven equalities follow from the same term-by-term comparison. Thus these conditions are not merely formal identities: they are exactly the equations that make the numerical and exact Taylor expansions coincide through fourth degree.

It remains to verify them for the coefficients in (3). Direct calculation gives

$$b^T \mathbf{1} = 1, \qquad b^T c = \frac{1}{2}, \qquad b^T c^2 = \frac{1}{3}, \qquad b^T c^3 = \frac{1}{4}, \tag{15}$$

and

$$Ac = \frac{1}{2} c^2, \qquad b^T A = b^T (I - C). \tag{16}$$

Therefore

$$\begin{aligned} b^T A c &= \frac{1}{2} b^T c^2 = \frac{1}{6}, & b^T C A c &= \frac{1}{2} b^T c^3 = \frac{1}{8}, \\ b^T A c^2 &= b^T (c^2 - c^3) = \frac{1}{12}, & b^T A^2 c &= \frac{1}{2} b^T A c^2 = \frac{1}{24}. \end{aligned} \tag{17}$$

Hence all conditions in (14) hold, and comparison of (9) with (13) proves the one-step defect estimate

$$\|y(t_n + h) - y_{n+1}\| = O(h^5). \tag{18}$$

Finally, let e_n be the global error. For a smooth Lipschitz vector field, the Runge–Kutta one-step map is Lipschitz with factor $1 + Lh$ for sufficiently small h. The local estimate (18) then gives

$$\|e_{n+1}\| \le (1 + Lh)\|e_n\| + C h^5. \tag{19}$$

For $nh \le T$, the discrete Gronwall estimate yields

$$\|e_n\| \le C h^5 \sum_{j=0}^{n-1} (1 + Lh)^j \le C_T h^4. \tag{20}$$

Thus the two-stage Gauss–Legendre method has global order four.

4 Exact Jacobian of the stage root

Introduce the stage field Jacobians and stage sensitivities

$$F_i := D_z f(t_n + c_i h, Z_i), \qquad S_i := \frac{\partial Z_i}{\partial z_n}. \tag{21}$$

Differentiating (4) with respect to z_n gives

$$S_i = I_{2d} + h \sum_{j=1}^{2} a_{ij} F_j S_j. \tag{22}$$

Equivalently,

$$\underbrace{\begin{pmatrix} I_{2d} - \frac{h}{4}F_1 & -h\left(\frac{1}{4} - r\right)F_2 \\ -h\left(\frac{1}{4} + r\right)F_1 & I_{2d} - \frac{h}{4}F_2 \end{pmatrix}}_{M = D_{(Z_1,Z_2)}R} \begin{pmatrix} S_1 \\ S_2 \end{pmatrix} = \begin{pmatrix} I_{2d} \\ I_{2d} \end{pmatrix}. \tag{23}$$

Here M is the Jacobian of the coupled stage residual. It is also the matrix used by an exact Newton linearization, but it is *not* the Jacobian of the physical one-step map.

Assuming that the converged stage root is regular, the implicit function theorem gives

$$\begin{pmatrix} S_1 \\ S_2 \end{pmatrix} = M^{-1} \begin{pmatrix} I_{2d} \\ I_{2d} \end{pmatrix}. \tag{24}$$

In an implementation, equation (23) should be solved directly; the inverse in (24) is only notation.

5 Exact Jacobian of the physical step

Define the Jacobian of the map (5) by

$$\mathcal{J}_n := D_{z_n} \Phi_{h,t_n}(z_n).$$

Differentiating the accepted update and using (21) yields

$$\boxed{\mathcal{J}_n = I_{2d} + h \sum_{i=1}^{2} b_i F_i S_i = I_{2d} + \frac{h}{2}\left(F_1 S_1 + F_2 S_2\right).} \tag{25}$$

Combining (24) and (25), the same Jacobian can be written as

$$\boxed{\mathcal{J}_n = I_{2d} + h \begin{pmatrix} \frac{1}{2}F_1 & \frac{1}{2}F_2 \end{pmatrix} M^{-1} \begin{pmatrix} I_{2d} \\ I_{2d} \end{pmatrix}.} \tag{26}$$

This is the exact tangent of the ideal converged root. It differentiates the mathematical Gauss–Legendre map, not a finite Newton stopping rule.

6 Direct proof of symplecticity with the step Jacobian

The one-step map is symplectic if and only if

$$\mathcal{J}_n^T \Omega \mathcal{J}_n = \Omega. \tag{27}$$

Set

$$K_i := F_i S_i. \tag{28}$$

Then equations (25) and (22) become

$$\mathcal{J}_n = I_{2d} + h \sum_i b_i K_i, \qquad I_{2d} = S_i - h \sum_j a_{ij} K_j. \tag{29}$$

Expanding the symplecticity defect gives

$$\begin{aligned} \mathcal{J}_n^T \Omega \mathcal{J}_n - \Omega = h \sum_i b_i \left(K_i^T \Omega + \Omega K_i \right) \\ + h^2 \sum_{i,j} b_i b_j K_i^T \Omega K_j. \end{aligned} \tag{30}$$

By the Hamiltonian identity (2),

$$\begin{aligned} K_i^T \Omega S_i + S_i^T \Omega K_i &= S_i^T \left(F_i^T \Omega + \Omega F_i \right) S_i \\ &= 0. \end{aligned} \tag{31}$$

Using the second identity in (29) on both occurrences of I_{2d} therefore gives

$$K_i^T \Omega + \Omega K_i = -h \sum_j a_{ij} \left(K_i^T \Omega K_j + K_j^T \Omega K_i \right). \tag{32}$$

Substitute (32) into (30). Renaming i and j in the second term arising from (32) produces

$$\boxed{\mathcal{J}_n^T \Omega \mathcal{J}_n - \Omega = h^2 \sum_{i,j} \left(b_i b_j - b_i a_{ij} - b_j a_{ji} \right) K_i^T \Omega K_j.} \tag{33}$$

Consequently, the defect vanishes whenever the Runge–Kutta coefficients obey

$$\boxed{b_i a_{ij} + b_j a_{ji} = b_i b_j, \qquad i, j = 1, 2.} \tag{34}$$

7 Verification for the Gauss–Legendre coefficients

For the diagonal terms in (34),

$$2b_1a_{11} - b_1^2 = 2\left(\frac{1}{2}\right)\left(\frac{1}{4}\right) - \frac{1}{4} = 0, \qquad 2b_2a_{22} - b_2^2 = 0. \tag{35}$$

For the off-diagonal terms,

$$\begin{aligned} b_1a_{12} + b_2a_{21} - b_1b_2 &= \frac{1}{2}\left(\frac{1}{4} - r\right) + \frac{1}{2}\left(\frac{1}{4} + r\right) - \frac{1}{4} \\ &= 0. \end{aligned} \tag{36}$$

The case $(i, j) = (2, 1)$ is the same equality with the two summands exchanged. Thus every coefficient multiplying $K_i^T \Omega K_j$ in (33) is zero, and hence

$$\boxed{\mathcal{J}_n^T \Omega \mathcal{J}_n = \Omega.} \tag{37}$$

Therefore the two-stage Gauss–Legendre method is symplectic when its implicit stage equations are solved exactly.

8 Symmetry and time reversibility

Symmetry is a property distinct from symplecticity. The method is symmetric when its backward step is exactly the inverse of its forward step. For the possibly time-dependent problem (1), this means

$$\boxed{\Phi_{-h,t_n+h} \circ \Phi_{h,t_n} = \mathrm{Id}\,.} \tag{38}$$

For an autonomous problem, equation (38) reduces to $\Phi_{-h} = \Phi_h^{-1}$.

For the forward step, write

$$f_1 = f(t_n + c_1 h, Z_1), \qquad f_2 = f(t_n + c_2 h, Z_2). \tag{39}$$

Equations (4) and (5) are then

$$\begin{aligned} Z_1 &= z_n + h(a_{11} f_1 + a_{12} f_2), \\ Z_2 &= z_n + h(a_{21} f_1 + a_{22} f_2), \qquad (40) \\ z_{n+1} &= z_n + h(b_1 f_1 + b_2 f_2). \qquad (41) \end{aligned}$$

Now start from z_{n+1} at time $t_n + h$ and apply the same method with step $-h$. Its stages $\widehat{Z}_1, \widehat{Z}_2$ satisfy

$$\widehat{Z}_i = z_{n+1} - h \sum_{j=1}^{2} a_{ij} f(t_n + h - c_j h, \widehat{Z}_j). \tag{42}$$

The Gauss nodes are reflected about the midpoint:

$$1 - c_1 = c_2, \qquad 1 - c_2 = c_1. \tag{43}$$

Thus the backward stage times are the forward stage times in reverse order. We shall verify that the backward stages are

$$\boxed{\widehat{Z}_1 = Z_2, \qquad \widehat{Z}_2 = Z_1.} \tag{44}$$

Under the identification (44), the first backward stage equation becomes

$$\begin{aligned} \widehat{Z}_1 &= z_{n+1} - h(a_{11} f_2 + a_{12} f_1) \\ &= z_n + h\left[(b_1 - a_{12}) f_1 + (b_2 - a_{11}) f_2\right]. \end{aligned} \tag{45}$$

For the coefficients (3),

$$b_1 - a_{12} = a_{21}, \qquad b_2 - a_{11} = a_{22}. \tag{46}$$

Consequently,

$$\widehat{Z}_1 = z_n + h(a_{21} f_1 + a_{22} f_2) = Z_2.$$

Likewise, the second backward stage satisfies

$$\begin{aligned} \widehat{Z}_2 &= z_{n+1} - h(a_{21} f_2 + a_{22} f_1) \\ &= z_n + h\left[(b_1 - a_{22}) f_1 + (b_2 - a_{21}) f_2\right]. \end{aligned} \tag{47}$$

Since

$$b_1 - a_{22} = a_{11}, \qquad b_2 - a_{21} = a_{12}, \tag{48}$$

we obtain

$$\widehat{Z}_2 = z_n + h(a_{11} f_1 + a_{12} f_2) = Z_1.$$

The proposed reversed stages therefore solve the coupled backward stage equations exactly.

The output of the backward step is now

$$\begin{aligned} \widehat{z}_n &= z_{n+1} - h(b_1 f_2 + b_2 f_1) \\ &= z_n + h(b_1 f_1 + b_2 f_2) - h(b_1 f_2 + b_2 f_1) \\ &= z_n, \end{aligned} \tag{49}$$

where the last equality uses $b_1 = b_2 = 1/2$. This proves (38).

More generally, an s-stage Runge–Kutta method is symmetric when its coefficients are invariant under stage reversal:

$$\boxed{c_i = 1 - c_{s+1-i}, \qquad b_i = b_{s+1-i}, \qquad a_{ij} = b_{s+1-j} - a_{s+1-i,s+1-j}.} \tag{50}$$

The coefficients in (3) satisfy all three identities for $s = 2$.

Differentiating the map identity (38) also gives the Jacobian form of reversibility:

$$\boxed{D\Phi_{-h,t_n+h}(z_{n+1})\, D\Phi_{h,t_n}(z_n) = I_{2d}.} \tag{51}$$

Thus the backward-step Jacobian is exactly the inverse of the forward-step Jacobian. As with symplecticity, a finite Newton tolerance and floating-point round-off can produce a small measured reversibility defect even though the ideal implicit method is exactly symmetric.

9 Implemented configurations

The public `GaussLegendre4` model uses the same collocation equations for every configuration. The selector `newton_jacobian_method` accepts `analytic`, `finite_difference`, or `auto`. Analytic mode uses exact guiding-centre particle blocks, finite-difference mode differentiates the coupled stage residual, and auto resolves the available dynamics capability. Absolute and relative tolerances and the maximum iteration count control the finite root solve.

With `track_energy=True`, the method also advances the time-conjugate momentum using the two converged Gauss stages and monitors the generalized Hamiltonian $K = H + k$. This triangular extension does not alter the physical collocation stages. These are configurations of one Gauss–Legendre model, not separate numerical methods.

10 Consequences and numerical interpretation

For one guiding-centre particle, $\mathcal{J}_n \in \mathbb{R}^{2\times 2}$ and

$$\mathcal{J}_n^T \Omega \mathcal{J}_n = \det(\mathcal{J}_n)\Omega.$$

Equation (37) therefore implies

$$\det(\mathcal{J}_n) = 1,$$

so the one-step map preserves oriented area. In dimensions greater than two, unit determinant is only a consequence of symplecticity, not an equivalent test.

If $\mathcal{J}_n$ denotes the local Jacobian at step n, the accumulated flow tangent after N steps is

$$\mathcal{P}_N = \mathcal{J}_{N-1} \cdots \mathcal{J}_1 \mathcal{J}_0.$$

Since a product of symplectic matrices is symplectic,

$$\mathcal{P}_N^T \Omega \mathcal{P}_N = \Omega.$$

The algebraic result concerns the exact root and the exact tangent (25). In floating-point computations, a finite Newton tolerance, round-off, or a finite-difference approximation of the map Jacobian produces a small nonzero measured defect

$$\left\| \mathcal{J}_n^T \Omega \mathcal{J}_n - \Omega \right\|.$$

That numerical floor does not contradict the exact symplecticity proved above.

References

- J. M. Sanz-Serna, "Runge–Kutta schemes for Hamiltonian systems," *BIT Numerical Mathematics*, 28 (1988), 877–883.
- E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration*, second edition, Springer, 2006.